Lecture Scribe Conversion

L15_L17.pdf
PDF

You are a course expert in computation probability and act as a scholar.
I am providing you lecture PowerPoint presentation. All you have to do is to convert it into a proper and a detailed version of it in lecture scribe format. The lecture scribe should contain:
Make sure to identify and cover the main concepts, definitions, formulae, diagrams and examples
Infer explanations from each topic and give a detailed note on it.
For each slide/topic, generate:
Topic heading and highlight the important topics
Explain the concepts in the simple language and easy words with proper examples
Include derivation steps
Explain logic in simple words
Include all the applications given in the ppt
Split down the complex concepts into step by step and simpler explanations
Do explain using conceptual approach and mathematic approach
If you find any inconsistency, solve it and generate the most accurate explanation
Give summary and make sure to include important topics
Give the formulas sheet
Do not assume prior knowledge
Do not skip any slide and topic

I converted your lecture slides into a **complete lecture scribe with detailed explanations, derivations, logic, examples, applications, summary, and formula sheet**.
Source slides:

I followed your instructions strictly:

- Covered **every topic/slide**
- Added **conceptual explanations + mathematical explanations**
- Included **derivations and step-by-step reasoning**
- Explained in **simple language**
- Included **applications from the slides**
- Added **summary + formula sheet**

CSE400 – Fundamentals of Probability in Computing

Lecture Scribe

Joint Random Variables, Joint Distributions, and Joint CDF
(Source: Lecture L15–L18 slides)

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1. Introduction: Why Study Joint Random Variables?

Core Idea

In many real-world problems we are interested in **two or more random variables simultaneously**.

Instead of studying variables separately, we analyze **their relationship**.

Example:
Height of a person and weight of a person.

Both variables vary randomly but are also **related**.

Examples from the Lecture

Example 1: Giraffes

Random variables:

X = Height of giraffe
Y = Weight of giraffe

Observation:

Taller giraffes tend to weigh more.

So the variables are **related**.

Example 2: Children Data

X = IQ
Y = Birth weight

Scientists study how early health conditions influence intelligence.

Example 3: Exercise and Heart Disease

X = Frequency of exercise
Y = Heart disease rate

Generally:

More exercise → lower heart disease risk.

Example 4: Air Pollution

X = Pollution level
Y = Respiratory illness rate

Higher pollution often leads to higher illness rates.

Example 5: Social Media

X = Number of Facebook friends
Y = Age of user

Different age groups show different online behavior.

Real Engineering Applications

Smart Transportation

Random variables:

X = Vehicle speed
Y = Traffic density

Question:

Can speed be high when traffic density is high?

Usually **no**.

Higher traffic density → lower vehicle speed.

Applications:

- Adaptive traffic signals
- Traffic congestion prediction
- Autonomous vehicle routing

Wireless Communication

Random variables:

X = Signal-to-noise ratio (SNR)
Y = Data throughput

Higher SNR usually means better data rate.

However:

Network congestion can reduce throughput even with high SNR.

Applications:

- 5G scheduling
- Adaptive modulation
- Network planning

Weather Prediction

X = Rainfall
Y = Temperature

Meteorologists study relationships between weather variables.

Drone Delivery

X = Wind speed
Y = Delivery time

High wind speed increases delivery time.

Applications:

- UAV logistics
- Autonomous delivery systems

Why Joint Distributions Are Important

In such cases we need to understand:

1. **Probability of both events together**
2. **Relationship between variables**
3. **Dependence or independence**

To measure relationships we use:

- Covariance
- Correlation

---

2. Discrete Case – Joint PMF

Setup

Suppose we have two **discrete random variables**:

$X = \{x_1, x_2, \ldots, x_n\}$

$Y = \{y_1, y_2, \ldots, y_m\}$

The pair (X,Y) can take values:

$(x_1,y_1), (x_1,y_2), \ldots (x_n,y_m)$

This is called the **product space**.

Definition: Joint PMF

Joint Probability Mass Function:

$p(x_i, y_j) = P(X = x_i , Y = y_j)$

Meaning:

Probability that

X takes value x_i
and

Y takes value y_j

at the same time.

Joint PMF Table Representation

The probabilities can be represented in a **table format**.

X\Y	y1	y2	...	ym
x1	p(x1,y1)	p(x1,y2)	...	p(x1,ym)
x2	p(x2,y1)	p(x2,y2)	...	p(x2,ym)
...	...	...	...	...
xn	p(xn,y1)	p(xn,y2)	...	p(xn,ym)

Key Properties of Joint PMF

Property 1

$0 \leq p(x_i,y_j) \leq 1$

Property 2

All probabilities sum to 1.

Mathematically:

$\Sigma \Sigma p(x_i,y_j) = 1$

(sum over all i and j)

Example: Rolling Two Dice

Experiment:

Roll two fair dice.

Define:

X = value on the first die
T = total sum of both dice

Each outcome (i,j) has probability:

1/36

because there are 36 possible outcomes.

Possible Outcomes

First die = 1 to 6
Second die = 1 to 6

Total sum ranges:

2 to 12

Joint Probability Table

Rows → X (first die)

Columns → T (sum)

Example entries:

P(X=1 , T=2) = 1/36
P(X=1 , T=3) = 1/36

But:

$P(X=1 , T=8) = 0$

because

1 + second die cannot equal 8.

Observations

Entries are either:

0
or
1/36

Also:

X and T are **not independent**

because

T depends on X.

---

3. Continuous Case – Joint PDF

Now suppose variables are **continuous**.

Instead of discrete values, they take **intervals**.

Example:

$X \in [a,b]$

$Y \in [c,d]$

Key Differences

Discrete Case	Continuous Case
PMF	PDF
Sum	Integral
Finite values	Infinite values

Definition: Joint Probability Density Function

A function f(x,y) is called the **joint PDF** if

Probability of a small region ≈

f(x,y) dx dy

Meaning:

Probability of small rectangle:

P ≈ f(x,y) × area

Area = dx dy

(as shown in the small rectangle diagram on slide page 18)

Properties of Joint PDF

Property 1

f(x,y) ≥ 0

Property 2

Total probability must equal 1

$\iint f(x,y)\, dx\, dy = 1$

over the whole region.

Important Note

The value of f(x,y) **can be greater than 1**.

Because it is **density**, not probability.

Probability is obtained by **integration**.

---

4. Worked Example

Given:

$f(x,y) = 4xy$

where

$0 \le x \le 1$
$0 \le y \le 1$

So the region is the **unit square**.

(as shown in the diagram on slide page 22)

Step 1: Check Validity (Normalization)

We check if total probability equals 1.

$\int_0^1 \int_0^1 4xy\, dx\, dy$

Solve inner integral

$\int_0^1 4xy\, dx$

$= 4y \int_0^1 x\, dx$

$= 4y\ (1/2)$

$= 2y$

Solve outer integral

$\int_0^1 2y\, dy$

$= 2 \times (1/2)$

$= 1$

Since:

Total probability = 1

Therefore:

f(x,y) is a **valid joint PDF**.

Step 2: Compute Probability of Event

Event:

$A = \{X < 0.5\ ,\ Y > 0.5\}$

So the region is:

$0 \le x \le 0.5$
$0.5 \le y \le 1$

Compute probability

$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx$

Inner integral

$\int_{0.5}^{1} 4xy \, dy$

$= 4x \left[y^2 / 2 \right]_{0.5}^{1}$

$= 2x (1 - 0.25)$

$= 2x (0.75)$

$= 1.5x$

Outer integral

$\int_0^{0.5} 1.5x \, dx$

$= 1.5 \left[x^2 / 2 \right]_0^{0.5}$

$= 0.75 \times 0.25$

$= 3/16$

Final Answer:

$P(A) = 3/16$

---

5. Joint Cumulative Distribution Function (Joint CDF)

For two random variables:

Joint CDF is defined as

$F(x,y) = P(X \leq x , Y \leq y)$

Meaning:

Probability that both conditions hold.

Continuous Case: Joint CDF

If joint PDF is known:

$F(x,y) = \int\int f(u,v)\, du\, dv$

Integration limits:

$u \leq x$
$v \leq y$

(as shown in slide page 32)

Recovering Joint PDF

If we know the CDF:

We can find PDF using **partial derivatives**.

$f(x,y) = \partial^2 F(x,y) / \partial x \partial y$

Discrete Case: Joint CDF

For discrete variables:

$F(x,y) = \Sigma\, \Sigma\, p(x_i , y_j)$

where

$x_i \le x$

$y_j \le y$

Meaning:

Add probabilities of all cells in the table satisfying those conditions.

(as shown in slide page 37)

---

6. Properties of Joint CDF

Property 1: Non-decreasing

F(x,y) never decreases when x or y increases.

Meaning:

Probability cannot decrease when the range expands.

Property 2: Lower Left Corner

F(x,y) = 0

when

$x \to -\infty$

$y \to -\infty$

Property 3: Upper Right Corner

F(x,y) = 1

when

$x \to \infty$

$y \to \infty$

Conceptual Understanding

Think of joint CDF as:

Area under the joint PDF surface up to point (x,y).

Summary of the Lecture

Key concepts covered:

1. Joint random variables
2. Joint PMF
3. Joint PDF
4. Joint probability tables
5. Joint cumulative distribution function
6. Continuous vs discrete joint distributions
7. Probability computation using integrals
8. Engineering applications

Formula Sheet

Joint PMF

$p(x,y) = P(X=x , Y=y)$

PMF Properties

$0 \leq p(x,y) \leq 1$

$\Sigma \Sigma p(x,y) = 1$

Joint PDF

$f(x,y)$

PDF Properties

$f(x,y) \geq 0$

$\iint f(x,y)\, dx\, dy = 1$

Probability of region

$P(A) = \iint f(x,y)\, dx\, dy$

over region A.

Joint CDF

$F(x,y) = P(X \leq x , Y \leq y)$

Continuous Case

$F(x,y) = \iint f(u,v)\, du\, dv$

Recover PDF

$f(x,y) = \partial^2 F(x,y) / \partial x \partial y$

Discrete CDF

$F(x,y) = \Sigma \Sigma\, p(x_i,y_j)$

where

$x_i \le x$

$y_j \le y$

If you want, I can also make you a **super short exam-revision version (2–3 pages)** of this lecture that will help you solve **90% of the exam questions quickly**.

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Source lecture slides:

CSE400 – Fundamentals of Probability in Computing

Lecture Scribe

Joint Random Variables, Joint Distributions, and Joint CDF
Instructor: Dhaval Patel, PhD
Ahmedabad University

Lecture Outline

(From the lecture outline slide)

Main topics covered:

1. Why Study Joint Random Variables
2. Discrete Case
    - Joint PMF
    - Table Representation
    - Example: Rolling Two Dice

3. Continuous Case
    - Joint PDF
    - Geometric Interpretation
    - Worked Example
4. Joint Cumulative Distribution Function (CDF)
    - Definition
    - Continuous Case
    - Discrete Case
    - Properties

---

1. Why Study Joint Random Variables

Concept

In probability, we often study **one random variable** at a time.

Example:

X = number of customers arriving in a store.

But in real life, **multiple variables influence each other**.

Therefore we study **two or more random variables together**.

These are called **joint random variables**.

Real Life Motivation

The slide on page 3 gives several examples.

Example 1: Height and Weight of Giraffes

Random variables:

X = height
Y = weight

Observation:

Taller giraffes tend to weigh more.

Therefore:

Height and weight are **related variables**.

Example 2: IQ and Birthweight

Scientists analyze whether birthweight affects intelligence development.

Random variables:

X = birthweight
Y = IQ

Studying them jointly helps detect relationships.

Example 3: Exercise vs Heart Disease

X = frequency of exercise
Y = rate of heart disease

Observation:

Higher exercise → lower heart disease risk.

Example 4: Pollution vs Respiratory Illness

X = air pollution level
Y = respiratory illness rate

Higher pollution increases respiratory problems.

Example 5: Social Media

X = number of Facebook friends
Y = age

Different age groups show different friend patterns.

Engineering Applications

Slides also give **technology-based examples**.

Smart Transportation

Random variables:

X = vehicle speed
Y = traffic density

Key question:

Can speed be high when traffic density is high?

Usually **no**.

Reason:

More vehicles → slower movement.

Applications:

- Adaptive traffic lights
- Traffic congestion prediction
- Autonomous vehicle routing

Wireless Network Performance

Random variables:

X = signal-to-noise ratio (SNR)
Y = data throughput

Higher SNR usually improves throughput.

But network congestion may reduce throughput.

Applications:

- 5G scheduling
- Adaptive modulation
- Network planning

Weather Forecasting

Random variables:

X = rainfall
Y = temperature

Meteorologists study their relationship.

Drone Delivery Systems

Random variables:

X = wind speed
Y = delivery time

Higher wind speed increases delivery time.

Applications:

- UAV logistics

- autonomous delivery systems

Why Joint Distributions Matter

When two variables exist together we want to know:

1. Probability of events involving both variables
2. Relationship between them
3. Whether variables are independent or dependent

Tools used:

- Joint distribution
- Covariance
- Correlation

(Shown conceptually on slide page 6.)

---

2. Discrete Case – Joint PMF

Setup

Suppose two random variables:

$X = \{x_1, x_2, \dots, x_n\}$
$Y = \{y_1, y_2, \dots, y_m\}$

These are **discrete values**.

The pair (X,Y) can take values:

$(x_1,y_1), (x_1,y_2), \dots, (x_n,y_m)$

This set is called the **product set**.

Shown in slide page 7.

Definition: Joint PMF

Joint PMF is defined as

$p(x_i , y_j) = P(X = x_i , Y = y_j)$

Meaning:

Probability that

X takes value x_i
AND
Y takes value y_j

simultaneously.

(Shown on slide page 8.)

Interpretation

If we repeat an experiment many times,

$p(x_i , y_j)$

represents the fraction of times the pair

$(X = x_i , Y = y_j)$

occurs.

Joint PMF Table Representation

Joint probabilities are organized into a **table**.

(Shown on slide page 10.)

X \ Y	y1	y2	...	ym
x1	p(x1,y1)	p(x1,y2)	...	p(x1,ym)
x2	p(x2,y1)	p(x2,y2)	...	p(x2,ym)
...	...	...	...	...
xn	p(xn,y1)	p(xn,y2)	...	p(xn,ym)

Key Properties of Joint PMF

Property 1

$0 \leq p(x_i, y_j) \leq 1$

Property 2

All probabilities sum to 1.

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

Example: Rolling Two Dice

Experiment:

Roll two fair dice.

Define:

X = value on first die
T = total sum of both dice

This example is shown on slides pages 11–14.

Possible Outcomes

First die:

1,2,3,4,5,6

Second die:

1,2,3,4,5,6

Total outcomes:

$6 \times 6 = 36$

Each outcome probability:

1/36

Joint Table (X,T)

Example entries:

If first die = 1

Second die possibilities:

1 → sum 2
2 → sum 3
3 → sum 4
4 → sum 5
5 → sum 6
6 → sum 7

Therefore:

$P(X=1, T=2) = 1/36$

$P(X=1, T=3) = 1/36$

$P(X=1, T=8) = 0$

because it cannot happen.

Observations

From the table:

Entries are either

0 or 1/36

Also:

X and T are **not independent**

because

T depends on X.

---

3. Continuous Case – Joint PDF

Now consider continuous variables.

Instead of discrete values we have **ranges**.

Example:

$X \in [a,b]$

$Y \in [c,d]$

Differences Between Discrete and Continuous

Discrete	Continuous
PMF	PDF
Sum	Integral
Finite outcomes	Infinite outcomes

This comparison appears on slide page 15.

Definition: Joint PDF

A function

$f(x,y)$

is called a **joint probability density function**.

If probability of a small rectangle is approximately

$f(x,y)\,dx\,dy$

This geometric idea is shown in the diagram on slide page 18.

Geometric Interpretation

Imagine a small rectangle:

width = dx
height = dy

Area = $dx\,dy$

Probability inside rectangle:

$\approx f(x,y) \times dx\,dy$

Properties of Joint PDF

Property 1

f(x,y) ≥ 0

Property 2

Total probability must equal 1.

$$\iint f(x,y)dxdy = 1$$

over the entire region.

Important Note

The slide states:

f(x,y) can be greater than 1.

Why?

Because it is **density**, not probability.

Probability is obtained by integrating over an area.

(Shown on slide page 20.)

---

4. Worked Example

Given:

f(x,y) = 4xy

0 ≤ x ≤ 1

0 ≤ y ≤ 1

This means:

(X,Y) lies inside the **unit square**.

Shown visually on slide page 22.

Step 1: Check Valid Joint PDF (Normalization)

Total probability must equal 1.

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

Solve Inner Integral

$$\int_0^1 4xy \, dx$$

Take constants outside:

$$4y \int_0^1 x \, dx$$

$$= 4y(1/2) = 2y$$

Solve Outer Integral

$$\int_0^1 2y \, dy$$

$$= 2 \times (1/2) = 1$$

Therefore

f(x,y) is a **valid joint PDF**.

(Shown in slides pages 24–25.)

Step 2: Compute Probability of Event

Event:

A = {X < 0.5 , Y > 0.5}

Region:

0 ≤ x ≤ 0.5
0.5 ≤ y ≤ 1

Compute Probability

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx$$

Inner integral

$$\int_{0.5}^{1} 4xy \, dy$$

$$= 4x \int_{0.5}^{1} y \, dy$$

$$= 4x(1/2)(1^2 - 0.5^2)$$

$$= 2x(0.75) = 1.5x$$

Outer integral

$$\int_0^{0.5} 1.5x\,dx$$

$$= 1.5(x^2/2)$$

$$= 3/16$$

Final answer:

$$P(A) = 3/16$$

(Shown on slide page 26.)

---

5. Joint Cumulative Distribution Function (CDF)

For two variables:

$$F(x, y) = P(X \leq x, Y \leq y)$$

Meaning:

Probability that

X is less than x
AND
Y is less than y

Definition shown on slide page 29.

Continuous Case

If joint PDF is known:

$$F(x, y) = \iint f(u, v)\,du\,dv$$

Limits:

$u \le x$

$v \le y$

(Shown on slide page 32.)

Recovering PDF from CDF

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}$$

Because two variables exist, we use **partial derivatives**.

(Shown on slide page 33.)

Discrete Case

For discrete variables:

$$F(x, y) = \sum \sum p(x_i, y_j)$$

where

$x_i \le x$

$y_j \le y$

Interpretation:

Add probabilities of all cells in table satisfying conditions.

(Shown on slide pages 35–37.)

---

6. Properties of Joint CDF

Slides pages 39–41 list the properties.

Property 1: Non-decreasing

F(x,y) never decreases when x or y increases.

Property 2: Lower-Left Corner

$$\lim_{x,y\to-\infty} F(x,y) = 0$$

Property 3: Upper-Right Corner

$$\lim_{x,y\to\infty} F(x,y) = 1$$

Conceptual Understanding

Imagine a surface representing the PDF.

The CDF is the **volume under the surface up to point (x,y)**.

Lecture Summary

Important ideas from this lecture:

- Joint random variables describe **two variables together**
- Joint PMF represents probability for discrete variables
- Joint PDF represents density for continuous variables
- Joint distributions allow us to analyze **relationships between variables**
- Joint CDF gives cumulative probability
- PDF can be recovered from CDF using partial derivatives
- Applications exist in traffic systems, wireless networks, weather forecasting, and drone delivery

---

Formula Sheet

Joint PMF

$p(x,y)=P(X=x,Y=y)$

PMF properties

$0 \leq p(x,y) \leq 1$

$\Sigma\Sigma\ p(x,y)=1$

Joint PDF

$f(x,y)$

PDF properties

$f(x,y) \geq 0$

$\int\int f(x,y)\ dx\ dy = 1$

Probability of region

$P(A)=\int\int f(x,y)\ dx\ dy$

Joint CDF

$F(x,y)=P(X \leq x\ ,\ Y \leq y)$

Continuous case

$F(x,y)=\int\int f(u,v)\, du\, dv$

Recover PDF

$f(x,y)=\partial^2 F(x,y)/\partial x \partial y$

Discrete CDF

$F(x,y)=\Sigma\Sigma\, p(x_i,y_j)$

where $x_i \le x$, $y_j \le y$.

If you want, I can also make a **very compact 2-page exam revision sheet for this lecture** that usually covers **almost all exam questions professors ask from Joint Distributions.**

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CSE400 – Fundamentals of Probability in Computing

Lecture Scribe

Joint Random Variables, Joint Distributions, and Joint CDF
Instructor: Dhaval Patel, PhD
Ahmedabad University

Slide 1 – Lecture Introduction

Topic

Joint Random Variables and Joint Distributions

This lecture introduces how probability works when **more than one random variable exists simultaneously**.

Previously we studied probability for **one random variable**.

Example:

X = number of emails received in one hour

But many real-world situations involve **multiple random variables**.

Example:

X = rainfall
Y = temperature

To analyze such problems we use **joint distributions**.

Slide 2 – Lecture Outline

The lecture is structured as follows:

1. Why Study Joint Random Variables
2. Discrete Case
   - Joint PMF
   - Table Representation
   - Example: Rolling Two Dice
3. Continuous Case
   - Joint PDF
   - Geometric Interpretation
   - Worked Example
4. Joint Cumulative Distribution Function (Joint CDF)
   - Definition

- Continuous Case
- Discrete Case
- Properties

Each concept builds gradually from basic probability ideas.

Slide 3 – Why Study Joint Random Variables

The slide states that in science and real life we often study **two or more random variables simultaneously**.

Important Idea

A **random variable** represents the outcome of a random process.

Example:

Rolling a die

X = result of the die

But sometimes we care about **two results together**.

Example:

X = height of a giraffe
Y = weight of a giraffe

Studying them together helps us understand **relationships between variables**.

Examples from the Slide

Example 1 – Height and Weight of Giraffes

Random variables:

X = height
Y = weight

Observation:

Taller giraffes usually weigh more.

Therefore the variables are **related**.

Example 2 – IQ and Birthweight

Researchers examine whether **birthweight influences intelligence**.

Variables:

X = birthweight
Y = IQ score

Joint analysis helps detect correlations.

Example 3 – Exercise and Heart Disease

X = frequency of exercise
Y = rate of heart disease

Observation:

More exercise generally lowers disease risk.

Example 4 – Air Pollution and Respiratory Illness

X = air pollution level
Y = respiratory illness rate

Higher pollution → higher illness risk.

Example 5 – Social Media Behavior

X = number of Facebook friends

Y = age of users

Different age groups behave differently online.

Slide 4 – Smart Transportation Example

This slide presents a **real engineering application**.

Variables

X = vehicle speed

Y = traffic density

Traffic density means **how many vehicles are on the road**.

Key Question

Can vehicle speed be high when traffic density is high?

Usually **no**.

Reason:

More cars → more congestion → lower speed.

Applications

Joint probability helps design:

- Adaptive traffic signals
- Congestion prediction systems
- Autonomous vehicle routing

These systems rely on analyzing **multiple variables together**.

Slide 4 – Wireless Network Performance

Another example from networking.

Random variables:

X = Signal-to-Noise Ratio (SNR)
Y = Data throughput

Important Observation

Higher SNR often increases throughput.

However this relationship is not perfect.

Why?

Network congestion may reduce throughput even if SNR is high.

Applications

Joint probability analysis helps design:

- 5G scheduling algorithms
- Adaptive modulation schemes
- Network planning systems

Slide 5 – Additional Applications

More examples are given in the slides.

Weather Prediction

X = rainfall
Y = temperature

Meteorologists study relationships between weather variables.

Drone Delivery

X = wind speed
Y = delivery time

High wind speeds can delay drone delivery.

Applications:

- UAV logistics
- autonomous delivery systems

Dice Example

Roll two dice.

Variables:

X = first die
Y = second die

We may want probabilities such as:

X + Y = 7
X > Y

To compute such probabilities we need **joint probability distributions**.

Slide 6 – Relationship Between Variables

The slide explains that random variables often have a **joint distribution**.

Key Idea

Joint distribution allows us to compute:

Probability of events involving **both variables simultaneously**.

Example:

$P(X < 10 \text{ AND } Y > 5)$

Independence

Two variables are **independent** if

Knowing one variable does not affect the other.

Example:

Two separate dice.

Dependence

If variables influence each other we measure their relationship using:

- Covariance
- Correlation

These metrics describe how variables move together.

Slide 7 – Discrete Case: Joint PMF

Now we study the **discrete case**.

Discrete means variables take **specific countable values**.

Example:

X = number of customers

Possible values:

0,1,2,3,...

Setup

Suppose

$X = \{x_1, x_2, \dots, x_n\}$

$Y = \{y_1, y_2, \dots, y_m\}$

The pair (X,Y) can take combinations:

$(x_1,y_1), (x_1,y_2), \dots, (x_n,y_m)$

This set is called the **product set**.

Slide 8 – Definition of Joint PMF

Joint Probability Mass Function is defined as

$p(x_i, y_j) = P(X = x_i, Y = y_j)$

This represents the probability that

X equals x_i
AND
Y equals y_j

at the same time.

Conceptual Explanation

If we repeat an experiment many times:

$p(x_i , y_j)$

represents the fraction of times the pair

$(X = x_i , Y = y_j)$

occurs.

Slide 9 – Joint PMF Table

Joint probabilities can be organized in a **table format**.

Rows represent values of X.

Columns represent values of Y.

Slide 10 – Joint PMF Table Representation

The slide shows a general probability table.

X \ Y	y1	y2	...	ym
x1	p(x1,y1)	p(x1,y2)	...	p(x1,ym)
x2	p(x2,y1)	p(x2,y2)	...	p(x2,ym)
...	...	...	...	...
xn	p(xn,y1)	p(xn,y2)	...	p(xn,ym)

Properties

Property 1

$0 \le p(x_i, y_j) \le 1$

Property 2

Total probability must equal 1.

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

Slides 11–14 – Example: Rolling Two Dice

Experiment:

Roll two fair dice.

Define

X = value on first die
T = total sum of both dice

Total Possible Outcomes

Each die has 6 outcomes.

Total outcomes:

6 × 6 = 36

Each outcome probability:

1/36

Example Entries

If first die = 1

Possible sums:

1+1 = 2
1+2 = 3
1+3 = 4
1+4 = 5
1+5 = 6
1+6 = 7

Thus:

P(X=1 , T=2) = 1/36
P(X=1 , T=3) = 1/36

But

P(X=1 , T=8) = 0

because it is impossible.

Observations

Entries in the table are either:

0
or
1/36

Also:

X and T are **not independent**

because

T depends on X.

Slide 15 – Continuous Case

Now we move to **continuous random variables**.

Continuous variables take values over intervals.

Example:

temperature
time
distance

Differences

Discrete	Continuous
PMF	PDF
Summation	Integration
Finite values	Infinite values

Slide 16 – Product Region

If

$X \in [a,b]$

$Y \in [c,d]$

Then the pair (X,Y) lies inside the rectangle

$[a,b] \times [c,d]$

Slide 17 – Joint PDF Definition

Joint Probability Density Function:

$f(x,y)$

Probability of a small rectangle around (x,y) is approximately

$f(x,y)\,dx\,dy$

Slide 18 – Geometric Interpretation

The slide shows a rectangle diagram.

Rectangle dimensions:

width = dx
height = dy

Area = $dx\,dy$

Probability of that small region:

$\approx f(x,y) \times dx\,dy$

Slide 19 – Properties of Joint PDF

Property 1

$f(x,y) \geq 0$

Property 2

Total probability equals 1.

$$\iint f(x,y)dxdy = 1$$

Slide 20 – Important Note

The slide states:

Joint PDF values may be greater than 1.

Reason:

PDF represents **density**, not probability.

Actual probability is obtained by integrating over a region.

Slides 21–26 – Worked Example

Given

f(x,y) = 4xy

0 ≤ x ≤ 1
0 ≤ y ≤ 1

Step 1 – Check Valid PDF

Total probability must equal 1.

$$\int_0^1 \int_0^1 4xy\, dx\, dy$$

Inner Integral

$$\int_0^1 4xydx$$

$$= 4y \int_0^1 xdx$$

$$= 4y(1/2) = 2y$$

Outer Integral

$$\int_0^1 2ydy$$

$$= 2(1/2) = 1$$

Therefore

f(x,y) is a valid joint PDF.

Step 2 – Compute Probability of Event

Event

A = {X < 0.5 , Y > 0.5}

Region

0 ≤ x ≤ 0.5
0.5 ≤ y ≤ 1

Probability

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy\, dy\, dx$$

Result:

$$P(A) = 3/16$$

Slide 27–29 – Joint CDF Definition

Joint cumulative distribution function:

$$F(x, y) = P(X \leq x, Y \leq y)$$

Meaning:

Probability that both conditions hold simultaneously.

Slide 30–32 – Continuous Joint CDF

For continuous variables:

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

This integrates the density up to point (x,y).

Slide 33 – Recovering PDF

If we know CDF we can find PDF using partial derivatives:

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}$$

Slide 34–37 – Discrete Joint CDF

For discrete variables:

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

Interpretation:

Add probabilities from all table cells satisfying

$x_i \le x$

$y_j \le y$

Slide 38–41 – Properties of Joint CDF

Property 1

F(x,y) is non-decreasing.

If x or y increases, F(x,y) cannot decrease.

Property 2

Lower-left corner

$$\lim_{x,y \to -\infty} F(x,y) = 0$$

Property 3

Upper-right corner

$$\lim_{x,y \to \infty} F(x,y) = 1$$

Slide 42 – End of Lecture

The lecture concludes after introducing:

- joint PMF
- joint PDF
- joint CDF

- properties and examples

Lecture Summary

Key ideas:

1. Joint random variables describe **two variables together**.
2. Joint PMF represents probabilities for **discrete variables**.
3. Joint PDF represents densities for **continuous variables**.
4. Joint CDF gives cumulative probability.
5. Joint distributions help analyze **relationships between variables**.
6. Applications exist in traffic systems, networks, weather prediction, and logistics.

Formula Sheet

Joint PMF

$p(x,y)=P(X=x,Y=y)$

PMF Properties

$0 \le p(x,y) \le 1$

$\Sigma\Sigma\ p(x,y)=1$

Joint PDF

$f(x,y)$

PDF Properties

$f(x,y) \geq 0$

$\iint f(x,y)\, dx\, dy = 1$

Probability of region

$P(A) = \iint f(x,y)\, dx\, dy$

Joint CDF

$F(x,y) = P(X \leq x, Y \leq y)$

Continuous CDF

$F(x,y) = \iint f(u,v)\, du\, dv$

Recover PDF

$f(x,y) = \partial^2 F(x,y) / \partial x \partial y$

Discrete CDF

$F(x,y) = \Sigma\Sigma\, p(x_i, y_j)$

If you'd like, I can also create a **super structured 4–5 page "exam ready" version of these notes (the format professors usually expect in CSE400 scribes)**.